MIPT & YSDA · AUTUMN 2026

Deep Generative Models

Lecture 7

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Recap of Previous Lecture

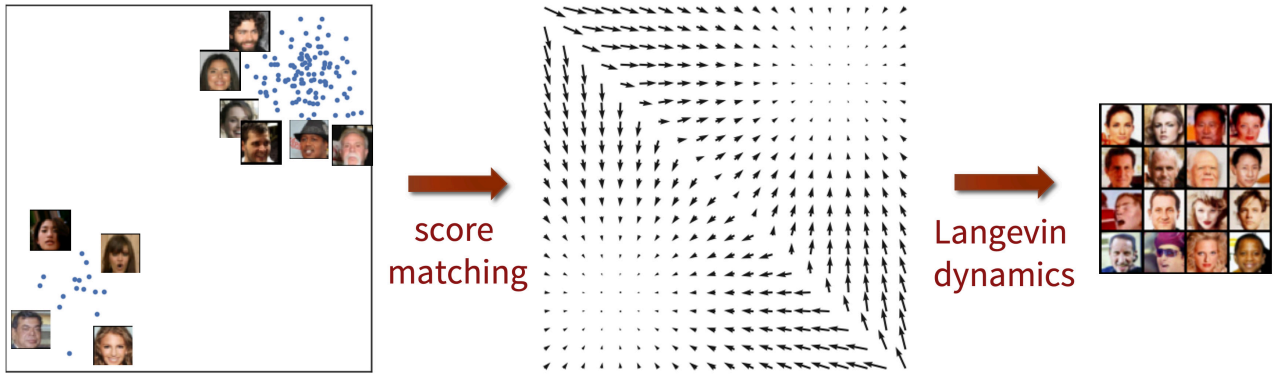
For an EBM $p_{\boldsymbol{\theta}}(\mathbf{x}) \propto e^{-E_{\boldsymbol{\theta}}(\mathbf{x})}$, its **score** is $\nabla_{\mathbf{x}} \log p_{\boldsymbol{\theta}}(\mathbf{x}) = -\nabla_{\mathbf{x}} E_{\boldsymbol{\theta}}(\mathbf{x})$.
We learn a vector network $\mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}) \approx \nabla_{\mathbf{x}} \log p_{\text{data}}(\mathbf{x})$ directly, then sample with noise.

Fisher Divergence

$$D_F(p_{\text{data}}, p_{\boldsymbol{\theta}}) = \frac{1}{2}\mathbb{E}_{p_{\text{data}}} \left\| \mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}) - \nabla_{\mathbf{x}} \log p_{\text{data}}(\mathbf{x}) \right\|_2^2 \to \min_{\boldsymbol{\theta}}$$

Langevin Dynamics

$$\mathbf{x}_{l+1} = \mathbf{x}_l + \frac{\eta}{2} \cdot \mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}_l) + \sqrt{\eta} \cdot \boldsymbol{\epsilon}_l, \quad \boldsymbol{\epsilon}_l \sim \mathcal{N}(0, \mathbf{I})$$



Recap of Previous Lecture

Let us perturb the original data with Gaussian noise $q(\mathbf{x}_\sigma|\mathbf{x}) = \mathcal{N}(\mathbf{x}, \sigma^2 \cdot \mathbf{I})$.

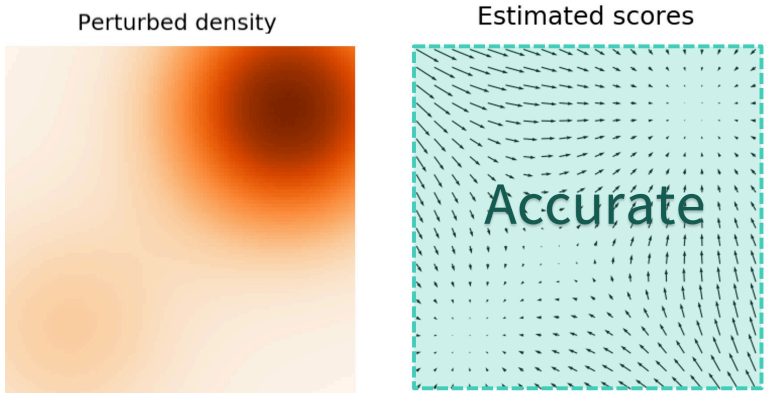
$$q(\mathbf{x}_\sigma) = \int q(\mathbf{x}_\sigma|\mathbf{x})p_{\text{data}}(\mathbf{x})d\mathbf{x}.$$

Assumption

The solution to

$$\frac{1}{2}\mathbb{E}_{q(\mathbf{x}_\sigma)}\left\|\mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}_\sigma}\log q(\mathbf{x}_\sigma)\right\|_2^2 \to \min_{\boldsymbol{\theta}}$$

satisfies $\mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) \approx \mathbf{s}_{\boldsymbol{\theta},0}(\mathbf{x}_0) = \mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x})$ if σ is sufficiently small.



Song Y. Generative Modeling by Estimating Gradients of the Data Distribution, blog post, 2021
Vincent P. A Connection Between Score Matching and Denoising Autoencoders, 2010

Recap of Previous Lecture

Theorem

Under mild regularity conditions, the last term is independent of $\boldsymbol{\theta}$:

$$\begin{aligned}&\mathbb{E}_{q(\mathbf{x}_\sigma)}\left\|\mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}_\sigma}\log q(\mathbf{x}_\sigma)\right\|_2^2 \\ &\quad= \mathbb{E}_{p_{\text{data}}(\mathbf{x})}\mathbb{E}_{q(\mathbf{x}_\sigma|\mathbf{x})}\left\|\mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}_\sigma}\log q(\mathbf{x}_\sigma|\mathbf{x})\right\|_2^2 + \text{const}\end{aligned}$$

Here,

$$\nabla_{\mathbf{x}_\sigma}\log q(\mathbf{x}_\sigma|\mathbf{x}) = -\frac{\mathbf{x}_\sigma - \mathbf{x}}{\sigma^2} = -\frac{\boldsymbol{\epsilon}}{\sigma}.$$

Tweedie's Formula

From Lecture 6, the MSE-optimal clean-data prediction is

$$\mathbb{E}_{q(\mathbf{x}|\mathbf{x}_\sigma)}[\mathbf{x}] = \mathbf{x}_\sigma + \sigma^2\nabla_{\mathbf{x}_\sigma}\log q(\mathbf{x}_\sigma).$$

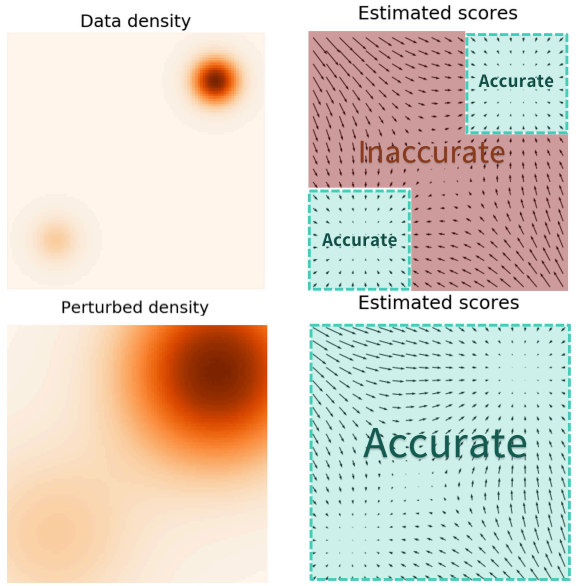
Vincent P. A Connection Between Score Matching and Denoising Autoencoders, 2010
Efron B. Tweedie's Formula and Selection Bias, 2011

Recap of Previous Lecture

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x})}\mathbb{E}_{\mathcal{N}(0,\mathbf{I})}\left\|\mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}+\sigma\boldsymbol{\epsilon})+\frac{\boldsymbol{\epsilon}}{\sigma}\right\|_2^2 \to \min_{\boldsymbol{\theta}}$$

$$\mathbf{x}_{l+1} = \mathbf{x}_l + \frac{\eta}{2}\cdot\mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_l) + \sqrt{\eta}\cdot\boldsymbol{\epsilon}_l$$

- For **small** σ, $\mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x})$ becomes inaccurate and Langevin dynamics fails to traverse modes
- For **large** σ, robustness in low-density regions is achieved, but the model learns a distribution that is overly corrupted



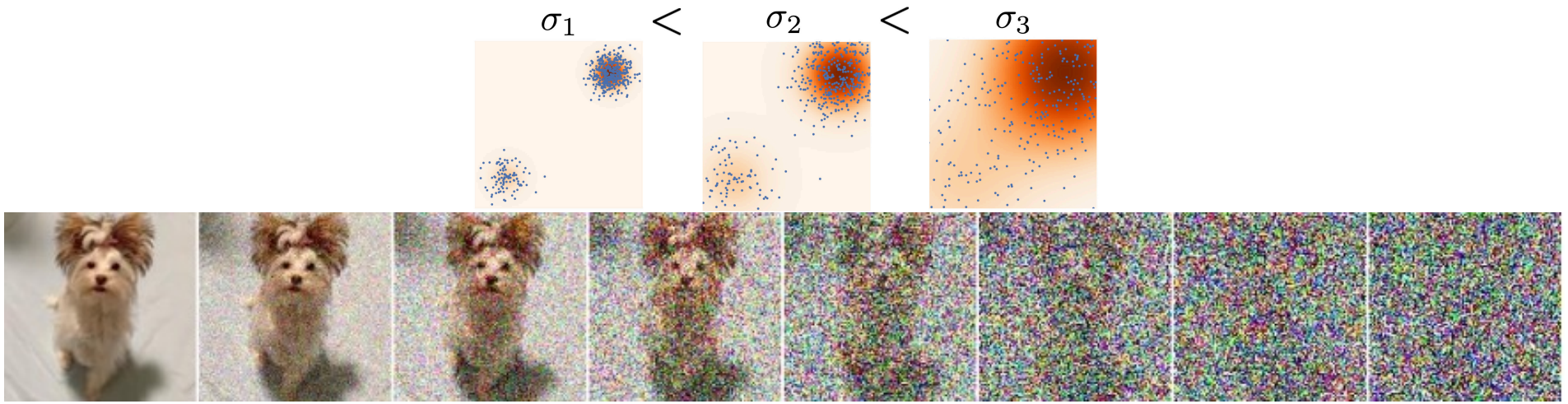
Recap of Previous Lecture

Noise-Conditioned Score Network (NCSN)

- Define a sequence of noise levels: $\sigma_1 < \sigma_2 < \cdots < \sigma_T$.
- Balance the score targets $-\boldsymbol{\epsilon}/\sigma_t$ using weights σ_t^2:

$$\sum_{t=1}^{T} \sigma_t^2 \cdot \mathbb{E}_{p_{\text{data}}(\mathbf{x})} \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x})} \left\| \mathbf{s}_{\boldsymbol{\theta},\sigma_t}(\mathbf{x}_t) - \nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t|\mathbf{x}) \right\|_2^2 \to \min_{\boldsymbol{\theta}}$$

- Sample using **annealed** Langevin dynamics (for $t = T, \ldots, 1$).



Song Y. et al. Generative Modeling by Estimating Gradients of the Data Distribution, 2019

Recap of Previous Lecture

Training

1. Sample $\mathbf{x}_0 \sim p_{\text{data}}(\mathbf{x})$, $t \sim U\{1, T\}$, $\boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I})$.
2. Compute noisy image $\mathbf{x}_t = \mathbf{x}_0 + \sigma_t \boldsymbol{\epsilon}$.
3. Compute loss $\mathcal{L} = \sigma_t^2 \left\| \mathbf{s}_{\boldsymbol{\theta}, \sigma_t}(\mathbf{x}_t) + \frac{\boldsymbol{\epsilon}}{\sigma_t} \right\|^2$.
4. Update $\boldsymbol{\theta}$ via stochastic gradient descent.

Sampling (Annealed Langevin Dynamics)

1. Sample $\mathbf{x}_T^0 \sim \mathcal{N}(0, \sigma_T^2 \mathbf{I}) \approx q(\mathbf{x}_T)$.
2. Update $\mathbf{x}_t^l$ via L steps of Langevin dynamics at each noise level σ_t:

$$\mathbf{x}_t^l = \mathbf{x}_t^{l-1} + \frac{\eta_t}{2} \mathbf{s}_{\boldsymbol{\theta}, \sigma_t}(\mathbf{x}_t^{l-1}) + \sqrt{\eta_t} \boldsymbol{\epsilon}_t^l.$$

3. Update $\mathbf{x}_{t-1}^0 := \mathbf{x}_t^L$ and proceed to σ_{t-1} if $t > 1$.

Song Y. et al. Generative Modeling by Estimating Gradients of the Data Distribution, 2019
Song Y. et al. Improved Techniques for Training Score-Based Generative Models, 2020

Outline

01 Forward Gaussian Diffusion Process

02 Reverse Gaussian Diffusion Process

03 Gaussian Diffusion Model as VAE

04 Diffusion ELBO Derivation

Outline

01 **Forward Gaussian Diffusion Process**

02 Reverse Gaussian Diffusion Process

03 Gaussian Diffusion Model as VAE

04 Diffusion ELBO Derivation

Forward Gaussian Diffusion Process

Let $\mathbf{x}_0 = \mathbf{x} \sim p_{\text{data}}(\mathbf{x})$, $\beta_t \ll 1$. Define a Markov chain:

$$\mathbf{x}_t = \sqrt{1-\beta_t}\mathbf{x}_{t-1} + \sqrt{\beta_t}\boldsymbol{\epsilon}_t, \quad \boldsymbol{\epsilon}_t \sim \mathcal{N}(0, \mathbf{I})$$

$$q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}(\sqrt{1-\beta_t}\mathbf{x}_{t-1}, \beta_t\mathbf{I})$$

Langevin Dynamics

$$\mathbf{x}_{l+1} = \mathbf{x}_l + \frac{\eta}{2} \cdot \nabla_{\mathbf{x}_l} \log p_{\boldsymbol{\theta}}(\mathbf{x}_l) + \sqrt{\eta}\boldsymbol{\epsilon}_l, \quad \boldsymbol{\epsilon}_l \sim \mathcal{N}(0, \mathbf{I})$$

$$\begin{aligned}
\mathbf{x}_t &= \sqrt{1-\beta_t}\,\mathbf{x}_{t-1} + \sqrt{\beta_t}\boldsymbol{\epsilon}_t \\
&\approx \left(1 - \frac{\beta_t}{2}\right)\mathbf{x}_{t-1} + \sqrt{\beta_t}\boldsymbol{\epsilon}_t \\
&= \mathbf{x}_{t-1} + \frac{\beta_t}{2}(-\mathbf{x}_{t-1}) + \sqrt{\beta_t}\boldsymbol{\epsilon}_t
\end{aligned}$$

From a Markov chain to Langevin dynamics: for small β_t, this Gaussian chain approximates LD targeting $\mathcal{N}(0, \mathbf{I})$, with step size $\eta = \beta_t$ and Gaussian score $-\mathbf{x}_{t-1}$.

Forward Diffusion: Sample at Any Timestep

Fix a schedule $0 < \beta_t < 1$. The noises $\boldsymbol{\epsilon}_t \sim \mathcal{N}(0, \mathbf{I})$ are independent across t and of $\mathbf{x}_0$.

$$\mathbf{x}_t = \sqrt{1-\beta_t}\mathbf{x}_{t-1} + \sqrt{\beta_t}\boldsymbol{\epsilon}_t, \qquad q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}(\sqrt{1-\beta_t}\mathbf{x}_{t-1}, \beta_t\mathbf{I}).$$

Exact Conditional Distribution

Let $\alpha_t = 1 - \beta_t$ and $\bar{\alpha}_t = \prod_{s=1}^{t} \alpha_s$. Then

$$q(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}(\sqrt{\bar{\alpha}_t}\,\mathbf{x}_0, (1-\bar{\alpha}_t)\mathbf{I}).$$

Sample $\mathbf{x}_t$ directly from $\mathbf{x}_0$, without simulating the intermediate steps:

$$\begin{aligned}
\mathbf{x}_t &= \sqrt{\alpha_t}\mathbf{x}_{t-1} + \sqrt{1-\alpha_t}\boldsymbol{\epsilon}_t \\
&= \sqrt{\alpha_t}\left(\sqrt{\alpha_{t-1}}\mathbf{x}_{t-2} + \sqrt{1-\alpha_{t-1}}\boldsymbol{\epsilon}_{t-1}\right) + \sqrt{1-\alpha_t}\boldsymbol{\epsilon}_t \\
&= \sqrt{\alpha_t\alpha_{t-1}}\mathbf{x}_{t-2} + (\sqrt{\alpha_t(1-\alpha_{t-1})}\boldsymbol{\epsilon}_{t-1} + \sqrt{1-\alpha_t}\boldsymbol{\epsilon}_t) \\
&= \sqrt{\alpha_t\alpha_{t-1}}\mathbf{x}_{t-2} + \sqrt{1-\alpha_t\alpha_{t-1}}\boldsymbol{\epsilon}'_t \\
&= \ldots = \sqrt{\bar{\alpha}_t}\,\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I}).
\end{aligned}$$

Noise Variance

Independent noises: variances add per coordinate.

$$\begin{aligned}
&\alpha_t(1-\alpha_{t-1}) + (1-\alpha_t) \\
&\quad = 1 - \alpha_t\alpha_{t-1}. \\
&\boldsymbol{\epsilon}'_t \sim \mathcal{N}(0, \mathbf{I}).
\end{aligned}$$

When Does Only Gaussian Noise Remain?

The closed-form conditional already tells us how the signal disappears:

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t}\,\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}, \qquad \boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I}), \quad \boldsymbol{\epsilon} \perp \mathbf{x}_0.$$

The Schedule Must Remove the Signal

For $0 < \beta_t < 1$, the required condition is

$$\bar{\alpha}_t = \prod_{s=1}^{t}(1-\beta_s) \longrightarrow 0 \quad \Longleftrightarrow \quad \sum_{t=1}^{\infty}\beta_t = \infty.$$

Many steps alone are not enough: $\beta_t = 2^{-t}$ leaves $\bar{\alpha}_\infty > 0$.

$$\sqrt{\bar{\alpha}_t} \to 0, \qquad \sqrt{1-\bar{\alpha}_t} \to 1 \quad \Longrightarrow \quad \mathbf{x}_t \xrightarrow{d} \mathcal{N}(0, \mathbf{I}).$$

The limit no longer depends on the initial data distribution p_{data}.

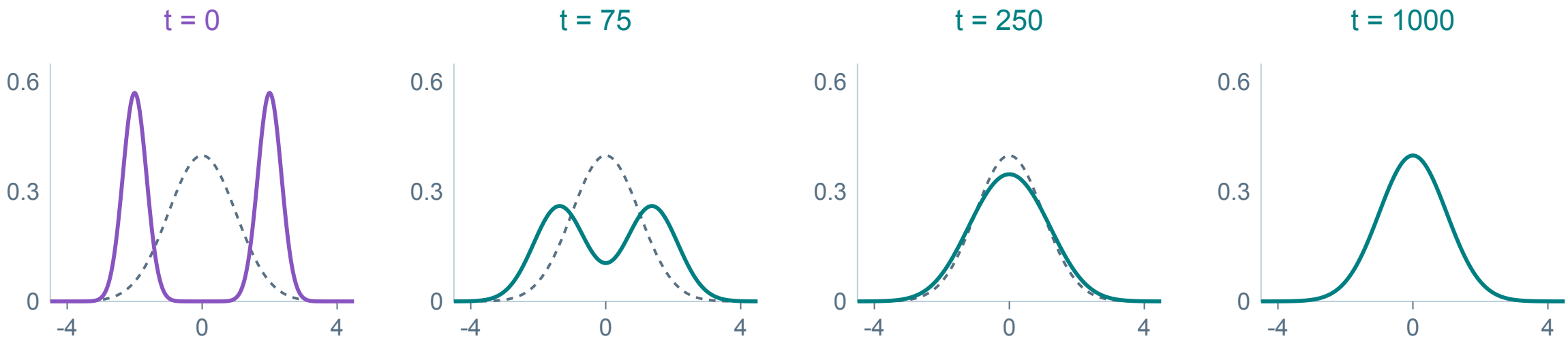
> **For finite T:** choose $\bar{\alpha}_T \ll 1$ with negligible residual signal $\sqrt{\bar{\alpha}_T}\mathbf{x}_0$.
> Then $q(\mathbf{x}_T) \approx \mathcal{N}(0, \mathbf{I})$: the terminal distribution is approximately standard Gaussian.

Forward Diffusion: From Data to Noise

$$q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}\left(\sqrt{1-\beta_t}\mathbf{x}_{t-1}, \beta_t\mathbf{I}\right);$$

$$q(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}\left(\sqrt{\bar{\alpha}_t}\mathbf{x}_0, (1-\bar{\alpha}_t)\mathbf{I}\right)$$

$$p_{\text{data}}(x_0) = \tfrac{1}{2}\mathcal{N}(-2, 0.35^2) + \tfrac{1}{2}\mathcal{N}(2, 0.35^2), \qquad \beta_t = 0.01$$



---- Standard Gaussian reference $\mathcal{N}(0,1)$.

The plots show marginal densities; the kernels above are conditional.

Denoising Score Matching

Let $\mathbf{x}_0 \sim p_{\text{data}}$, $\boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I})$, with $\boldsymbol{\epsilon}$ independent of $\mathbf{x}_0$.

NCSN

$$\mathbf{x}_t = \mathbf{x}_0 + \sigma_t \boldsymbol{\epsilon}$$

$$q(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}(\mathbf{x}_0, \sigma_t^2 \mathbf{I})$$

$$\nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t|\mathbf{x}_0) = -\frac{\mathbf{x}_t - \mathbf{x}_0}{\sigma_t^2}$$

$$q(\mathbf{x}_1) \approx p_{\text{data}}, \quad q(\mathbf{x}_T) \approx \mathcal{N}(0, \sigma_T^2 \mathbf{I})$$

Gaussian Diffusion

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t}\boldsymbol{\epsilon}$$

$$q(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}(\sqrt{\bar{\alpha}_t}\mathbf{x}_0, (1 - \bar{\alpha}_t)\mathbf{I})$$

$$\nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t|\mathbf{x}_0) = -\frac{\mathbf{x}_t - \sqrt{\bar{\alpha}_t}\mathbf{x}_0}{1 - \bar{\alpha}_t}$$

$$q(\mathbf{x}_1) \approx p_{\text{data}}, \quad q(\mathbf{x}_T) \approx \mathcal{N}(0, \mathbf{I})$$

Theorem (Denoising Score Matching)

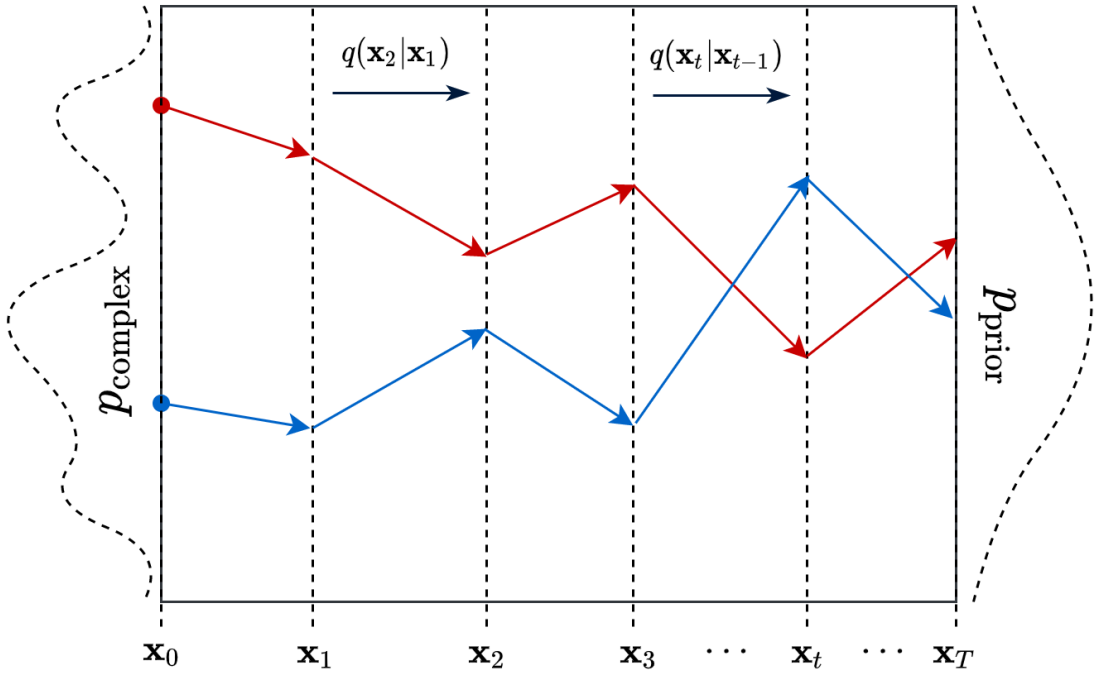
$$\mathbb{E}_{q(\mathbf{x}_t)} \left\| \mathbf{s}_{\boldsymbol{\theta},t}(\mathbf{x}_t) - \nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t) \right\|_2^2$$
$$= \mathbb{E}_{p_{\text{data}}(\mathbf{x}_0)} \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} \left\| \mathbf{s}_{\boldsymbol{\theta},t}(\mathbf{x}_t) - \nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t|\mathbf{x}_0) \right\|_2^2 + \text{const}$$

Note: Annealed Langevin dynamics applies to diffusion, too.
NCSN with annealed Langevin dynamics is an early score-based generative framework (2019–2020).

Song Y. et al. Generative Modeling by Estimating Gradients of the Data Distribution, 2019
Sohl-Dickstein J. Deep Unsupervised Learning using Nonequilibrium Thermodynamics, 2015

Forward Gaussian Diffusion Process

Diffusion describes the migration of particles from regions of high density to those of low density.



1. $\mathbf{x}_0 = \mathbf{x} \sim p_{\text{data}}(\mathbf{x})$
2. $\mathbf{x}_t = \sqrt{1-\beta_t}\mathbf{x}_{t-1} + \sqrt{\beta_t}\boldsymbol{\epsilon}_t,$
   $\boldsymbol{\epsilon}_t \sim \mathcal{N}(0, \mathbf{I}),\ t \geq 1$
3. Choose $\bar{\alpha}_T \ll 1$ to suppress $\sqrt{\bar{\alpha}_T}\mathbf{x}_0$.
   Then $q(\mathbf{x}_T) \approx \mathcal{N}(0, \mathbf{I})$.

If this process can be reversed, we can sample from $p_{\text{data}}(\mathbf{x})$ by starting from noise $\mathcal{N}(0, \mathbf{I}) \approx q(\mathbf{x}_T)$.
Our goal now becomes inverting this diffusion.

Outline

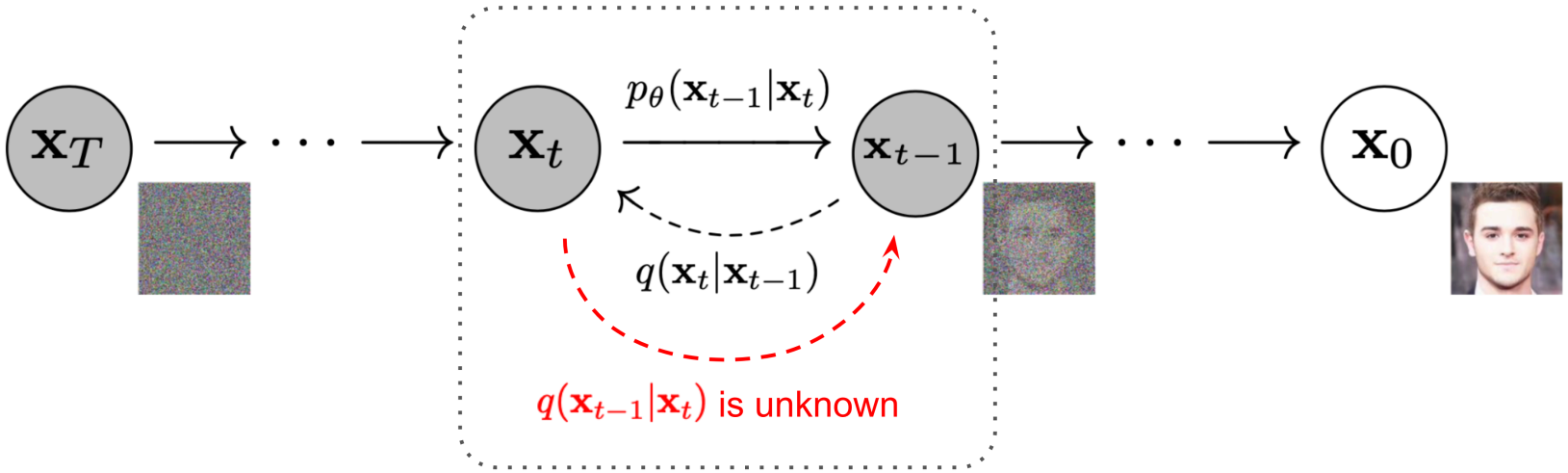
01 Forward Gaussian Diffusion Process

02 Reverse Gaussian Diffusion Process

03 Gaussian Diffusion Model as VAE

04 Diffusion ELBO Derivation

Reverse Gaussian Diffusion Process



Forward Process $$q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}\left(\sqrt{1-\beta_t}\mathbf{x}_{t-1}, \beta_t\mathbf{I}\right)$$

Reverse Process $$q(\mathbf{x}_{t-1}|\mathbf{x}_t) = \frac{q(\mathbf{x}_t|\mathbf{x}_{t-1})q(\mathbf{x}_{t-1})}{q(\mathbf{x}_t)} \approx p_{\boldsymbol{\theta}}(\mathbf{x}_{t-1}|\mathbf{x}_t)$$

$q(\mathbf{x}_{t-1})$ and $q(\mathbf{x}_t)$ are intractable:

$$q(\mathbf{x}_t) = \int q(\mathbf{x}_t|\mathbf{x}_0)p_{\text{data}}(\mathbf{x}_0)d\mathbf{x}_0$$

Weng L. What are Diffusion Models?, blog post, 2021

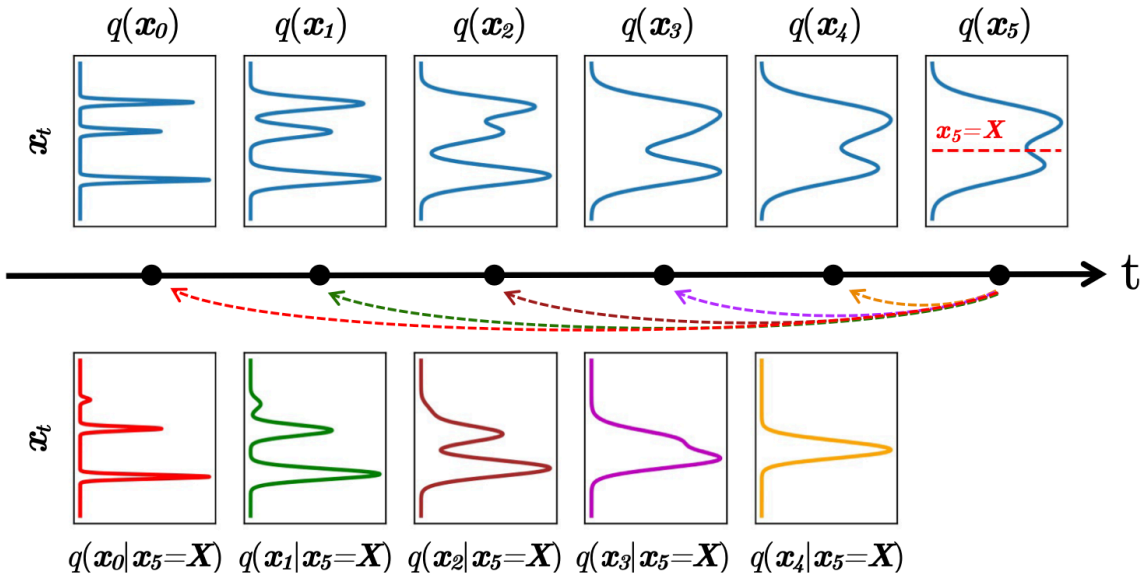
Reverse Gaussian Diffusion Process

Theorem (Feller, 1949)

If β_t is sufficiently small, the reverse conditional is approximately Gaussian:

$$q(\mathbf{x}_{t-1}|\mathbf{x}_t) = \frac{q(\mathbf{x}_t|\mathbf{x}_{t-1})q(\mathbf{x}_{t-1})}{q(\mathbf{x}_t)}.$$

The original DDPM used $T = 1000$ steps. Faster sampling methods will be discussed later.



Same Noisy Endpoint

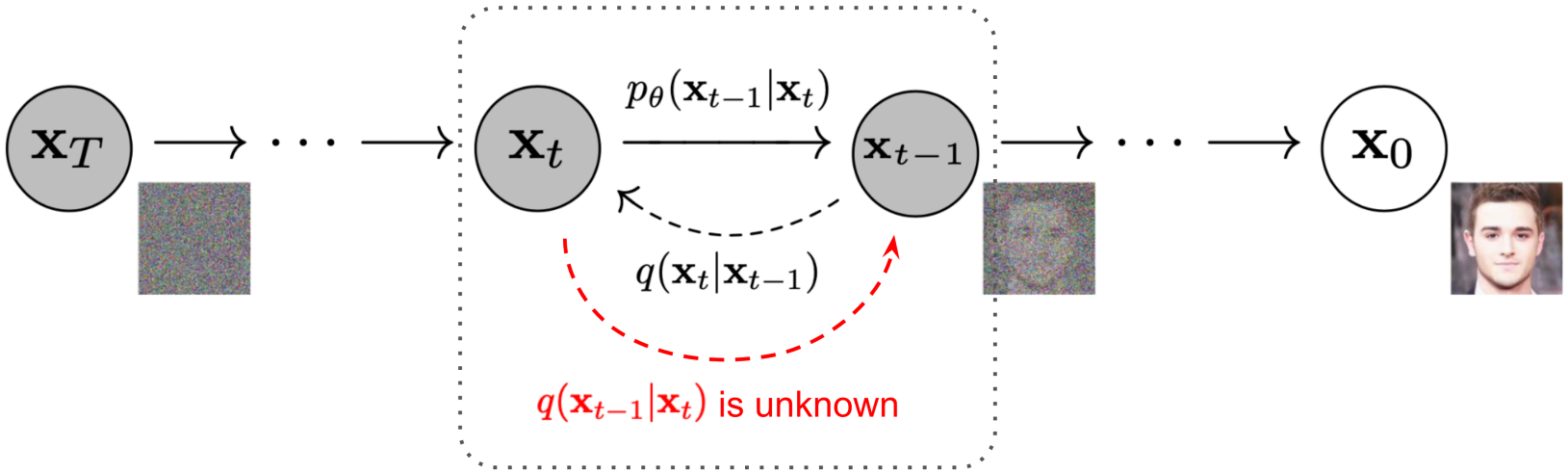
Top: marginals $q(x_s)$.
Bottom: posteriors for fixed $x_5 = X$.

$$q(x_s|x_5 = X) \propto q(x_5 = X|x_s)q(x_s).$$

- **Short reverse step:** nearly Gaussian.
- **Long reverse jump:** can be multimodal.

Feller W. On the theory of stochastic processes, with particular reference to applications, 1949
Ho J. et al. Denoising Diffusion Probabilistic Models, 2020
Xiao Z., Kreis K., Vahdat A. Tackling the generative learning trilemma with denoising diffusion GANs, 2021

Reverse Gaussian Diffusion Process (Ancestral Sampling)

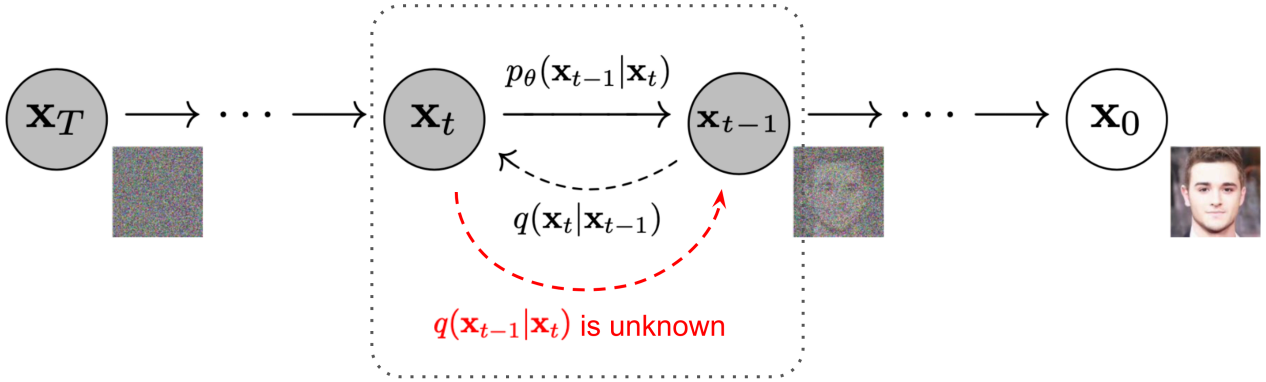


Forward Process $$q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}\left(\sqrt{1-\beta_t}\mathbf{x}_{t-1}, \beta_t\mathbf{I}\right)$$

Reverse Process $$q(\mathbf{x}_{t-1}|\mathbf{x}_t) \approx p_{\boldsymbol{\theta}}(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}\left(\boldsymbol{\mu}_{\boldsymbol{\theta},t}(\mathbf{x}_t), \boldsymbol{\sigma}^2_{\boldsymbol{\theta},t}(\mathbf{x}_t)\right)$$

If β_t is sufficiently small, $q(\mathbf{x}_{t-1}|\mathbf{x}_t)$ is approximately Gaussian.

Reverse Gaussian Diffusion Process (Ancestral Sampling)



Forward Process

$$q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}\left(\sqrt{1-\beta_t}\mathbf{x}_{t-1}, \beta_t\mathbf{I}\right)$$

1. $\mathbf{x}_0 \sim p_{\text{data}}(\mathbf{x}_0)$
2. $\mathbf{x}_t = \sqrt{1-\beta_t}\mathbf{x}_{t-1} + \sqrt{\beta_t}\boldsymbol{\epsilon}_t, \quad \boldsymbol{\epsilon}_t \sim \mathcal{N}(0, \mathbf{I})$
3. $q(\mathbf{x}_T) \approx \mathcal{N}(0, \mathbf{I})$

Reverse Process

$$p_{\boldsymbol{\theta}}(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}\left(\boldsymbol{\mu}_{\boldsymbol{\theta},t}(\mathbf{x}_t), \boldsymbol{\sigma}^2_{\boldsymbol{\theta},t}(\mathbf{x}_t)\right)$$

1. $\mathbf{x}_T \sim \mathcal{N}(0, \mathbf{I})$
2. $\mathbf{x}_{t-1} = \boldsymbol{\mu}_{\boldsymbol{\theta},t}(\mathbf{x}_t) + \boldsymbol{\sigma}_{\boldsymbol{\theta},t}(\mathbf{x}_t)\boldsymbol{\epsilon}_t, \quad \boldsymbol{\epsilon}_t \sim \mathcal{N}(0, \mathbf{I})$
3. $\mathbf{x}_0 \sim p_{\boldsymbol{\theta}}(\mathbf{x}_0) \approx p_{\text{data}}(\mathbf{x}_0)$

> **Note:** The forward process has no trainable parameters.

Conditioned Reverse Distribution

Reverse Kernel (Intractable)

$$q(\mathbf{x}_{t-1}|\mathbf{x}_t) = \frac{q(\mathbf{x}_t|\mathbf{x}_{t-1})q(\mathbf{x}_{t-1})}{q(\mathbf{x}_t)} \approx \mathcal{N}\big(\boldsymbol{\mu}_{\boldsymbol{\theta},t}(\mathbf{x}_t), \boldsymbol{\sigma}^2_{\boldsymbol{\theta},t}(\mathbf{x}_t)\big), \quad \beta_t \ll 1.$$

Conditioned Reverse Kernel (Tractable)

During training, $\mathbf{x}_0$ is known: substitute Gaussians and complete the square in $\mathbf{x}_{t-1}$.

$$\begin{aligned} q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) &= \frac{q(\mathbf{x}_t|\mathbf{x}_{t-1})q(\mathbf{x}_{t-1}|\mathbf{x}_0)}{q(\mathbf{x}_t|\mathbf{x}_0)} \\ &= \frac{\mathcal{N}(\sqrt{\alpha_t}\mathbf{x}_{t-1}, (1-\alpha_t)\mathbf{I})\,\mathcal{N}(\sqrt{\bar{\alpha}_{t-1}}\mathbf{x}_0, (1-\bar{\alpha}_{t-1})\mathbf{I})}{\mathcal{N}(\sqrt{\bar{\alpha}_t}\mathbf{x}_0, (1-\bar{\alpha}_t)\mathbf{I})} \\ &= \mathcal{N}\big(\tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t, \mathbf{x}_0), \tilde{\beta}_t\mathbf{I}\big). \end{aligned}$$

$$\tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t, \mathbf{x}_0) = \frac{\sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t} \cdot \mathbf{x}_t + \frac{\sqrt{\bar{\alpha}_{t-1}}(1-\alpha_t)}{1-\bar{\alpha}_t} \cdot \mathbf{x}_0$$

$$\tilde{\beta}_t = \frac{(1-\alpha_t)(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t} = \text{const}$$

What Does Knowing the Clean Data Change?

Same observation $x_t = 0$: without and with the clean value $x_0 = -2$.

Two possible clean values

$\mathbb{P}(x_0 = -2) = \mathbb{P}(x_0 = 2) = \frac{1}{2}$

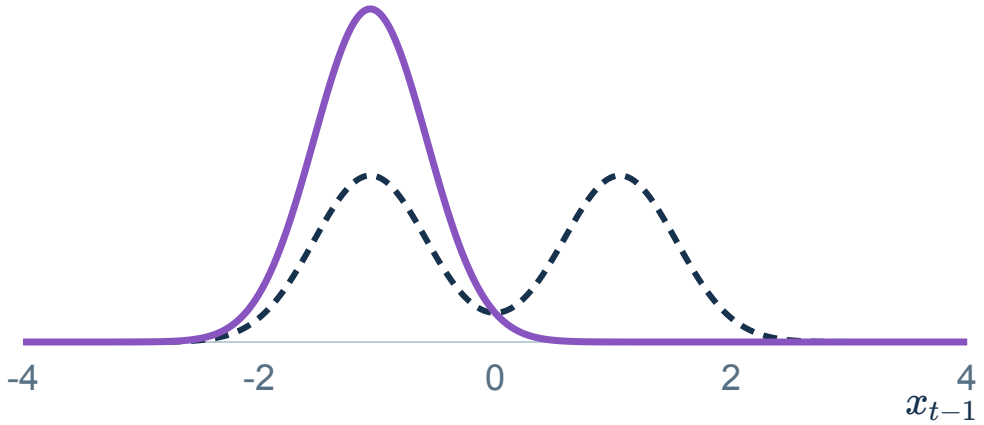
$x_0 = -2$ — $q(x_0 = -2|x_t)$ = 50.0%

$x_0 = 2$ — $q(x_0 = 2|x_t)$ = 50.0%

How plausible is each clean value after observing x_t?
These probabilities weight the mixture.

Exact toy example: $\alpha_t = 0.6,\ \bar{\alpha}_{t-1} = 0.65.$

Distribution of the previous state x_{t-1}



Observe only x_t

--- $q(x_{t-1}|x_t)$

Mixture over possible x_0

Also know $x_0 = -2$

— $q(x_{t-1}|x_t, x_0 = -2)$

One exact Gaussian

$$q(x_{t-1}|x_t) = \sum_{x_0 \in \{-2,2\}} q(x_{t-1}|x_t, x_0)\, q(x_0|x_t)$$

Knowing x_0 removes ambiguity about the clean value; uncertainty about x_{t-1} remains.

Distribution Summary

Forward process: with a suitable noise schedule, $q(\mathbf{x}_T) \approx \mathcal{N}(0, \mathbf{I})$ as the residual signal becomes negligible:

$$q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}(\sqrt{1-\beta_t} \cdot \mathbf{x}_{t-1}, \beta_t \cdot \mathbf{I});$$
$$q(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}(\sqrt{\bar{\alpha}_t} \cdot \mathbf{x}_0, (1-\bar{\alpha}_t) \cdot \mathbf{I}).$$

Reverse process refers to an intractable distribution that can be approximated by a normal distribution (with unknown parameters) for small β_t:

$$q(\mathbf{x}_{t-1}|\mathbf{x}_t) = \frac{q(\mathbf{x}_t|\mathbf{x}_{t-1})q(\mathbf{x}_{t-1})}{q(\mathbf{x}_t)} \approx \mathcal{N}\left(\boldsymbol{\mu}_{\boldsymbol{\theta},t}(\mathbf{x}_t), \boldsymbol{\sigma}^2_{\boldsymbol{\theta},t}(\mathbf{x}_t)\right)$$

Conditioned reverse process is a normal distribution with known parameters, describing how to denoise a noisy image $\mathbf{x}_t$ when we know the clean image $\mathbf{x}_0$.

$$q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \mathcal{N}(\tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t, \mathbf{x}_0), \tilde{\beta}_t \cdot \mathbf{I})$$

Noise Schedule through Signal-to-Noise Ratio

Signal and Noise

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}.$$

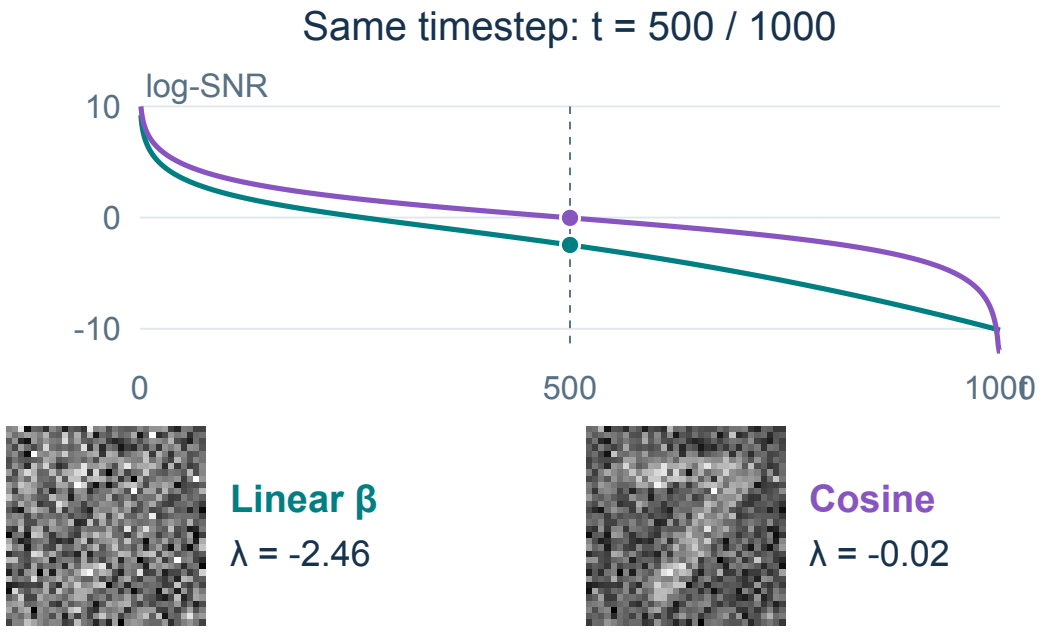
For $\boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I})$, compare squared scales:

$$\mathrm{SNR}(t) = \frac{\bar{\alpha}_t}{1-\bar{\alpha}_t}.$$

A Single Noise-Level Coordinate

$$\lambda_t = \log \mathrm{SNR}(t), \qquad \bar{\alpha}_t = \frac{1}{1+e^{-\lambda_t}}.$$

$\lambda_t > 0$: signal dominates.
$\lambda_t = 0$: equal signal and noise scales.
$\lambda_t < 0$: noise dominates.



Same clean image and Gaussian noise.
Fixed display scale; log-SNR axis clipped to [−12, 10].

Same timestep, different noise level.

Higher resolution: averaging suppresses independent noise while preserving coarse structure. Lower log-SNR helps corrupt this structure too.

Kingma D. et al. Variational Diffusion Models, 2021
Nichol A., Dhariwal P. Improved Denoising Diffusion Probabilistic Models, 2021
Hoogeboom E. et al. Simple diffusion: End-to-end diffusion for high resolution images, 2023

Outline

01 Forward Gaussian Diffusion Process

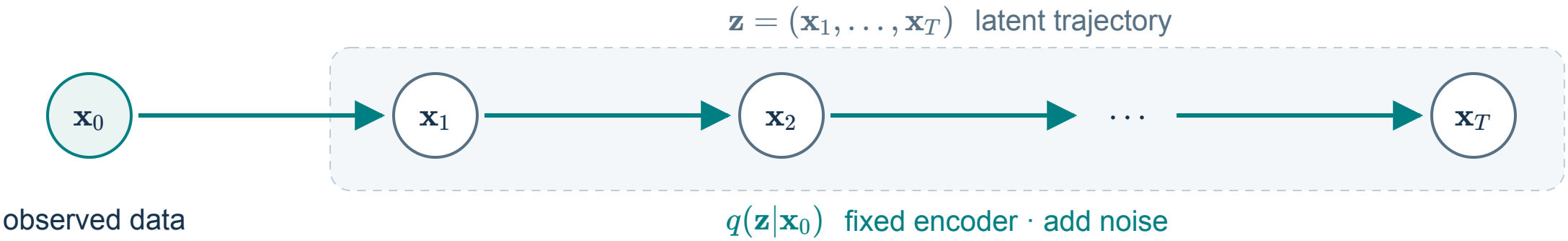
02 Reverse Gaussian Diffusion Process

03 Gaussian Diffusion Model as VAE

04 Diffusion ELBO Derivation

Diffusion as a VAE: Latent Variables and Encoder

$$\underbrace{\mathbf{x} = \mathbf{x}_0}_{\text{observed data}}, \qquad \underbrace{\mathbf{z} = (\mathbf{x}_1, \ldots, \mathbf{x}_T)}_{\text{entire latent trajectory}}.$$



Each $\mathbf{x}_t$ has the same dimension as $\mathbf{x}_0$; $\mathbf{z}$ contains **all T noisy states**.

Forward Diffusion = Fixed Encoder

Given a clean image, sample its noisy trajectory:

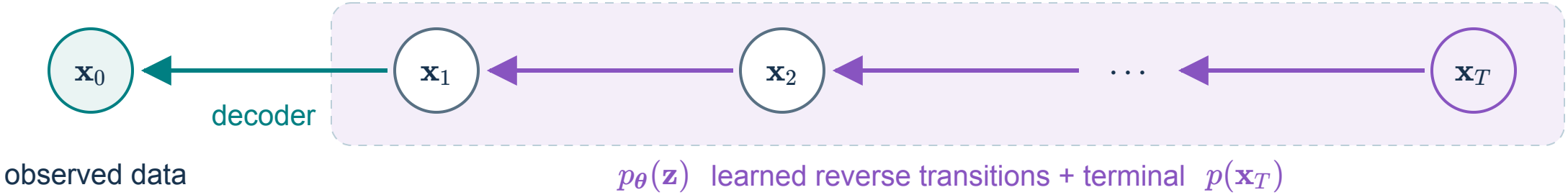
$$q(\mathbf{z}|\mathbf{x}) = q(\mathbf{x}_1, \ldots, \mathbf{x}_T|\mathbf{x}_0) = \prod_{t=1}^{T} q(\mathbf{x}_t|\mathbf{x}_{t-1}).$$

> **No trainable parameters:** the noise schedule fixes the variational posterior $q(\mathbf{z}|\mathbf{x})$.

Diffusion as a VAE: Prior and Decoder

$$p_{\boldsymbol{\theta}}(\mathbf{x}, \mathbf{z}) = p_{\boldsymbol{\theta}}(\mathbf{x}_0, \ldots, \mathbf{x}_T) = \underbrace{p_{\boldsymbol{\theta}}(\mathbf{x}_0|\mathbf{x}_1)}_{\text{decoder } p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{z})} \underbrace{p(\mathbf{x}_T) \prod_{t=2}^{T} p_{\boldsymbol{\theta}}(\mathbf{x}_{t-1}|\mathbf{x}_t)}_{\text{prior } p_{\boldsymbol{\theta}}(\mathbf{z})}.$$

$\mathbf{z} = (\mathbf{x}_1, \ldots, \mathbf{x}_T)$ latent trajectory



1. Start from noise: $p(\mathbf{x}_T) = \mathcal{N}(0, \mathbf{I})$.

2. Sample the latent trajectory: $\mathbf{x}_T \to \cdots \to \mathbf{x}_1$, completing the **prior** $p_{\boldsymbol{\theta}}(\mathbf{z})$.

3. Generate the observation: $\mathbf{x}_1 \to \mathbf{x}_0$ is the **decoder**.

By the Markov property, $p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{z}) = p_{\boldsymbol{\theta}}(\mathbf{x}_0|\mathbf{x}_1)$.

> **Fixed encoder, learned Markov-chain prior, one-step decoder.**

Outline

01 Forward Gaussian Diffusion Process

02 Reverse Gaussian Diffusion Process

03 Gaussian Diffusion Model as VAE

04 Diffusion ELBO Derivation

ELBO for Gaussian Diffusion Model

Standard ELBO

$$\log p_{\boldsymbol{\theta}}(\mathbf{x}) \geq \mathbb{E}_{q(\mathbf{z}|\mathbf{x})} \log \frac{p_{\boldsymbol{\theta}}(\mathbf{x}, \mathbf{z})}{q(\mathbf{z}|\mathbf{x})} = \mathcal{L}_{\boldsymbol{\phi},\boldsymbol{\theta}}(\mathbf{x}) \to \max_{\boldsymbol{\phi},\boldsymbol{\theta}}$$

Derivation

$$\begin{aligned}\mathcal{L}_{\boldsymbol{\phi},\boldsymbol{\theta}}(\mathbf{x}) &= \mathbb{E}_{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} \log \frac{p_{\boldsymbol{\theta}}(\mathbf{x}_0, \mathbf{x}_{1:T})}{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} \\ &= \mathbb{E}_{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} \log \frac{p(\mathbf{x}_T) \prod_{t=1}^{T} p_{\boldsymbol{\theta}}(\mathbf{x}_{t-1}|\mathbf{x}_t)}{\prod_{t=1}^{T} q(\mathbf{x}_t|\mathbf{x}_{t-1})}\end{aligned}$$

- Let's try to decompose the ELBO into individual KL divergence terms.
- We need to replace $q(\mathbf{x}_t|\mathbf{x}_{t-1})$ with $q(\mathbf{x}_{t-1}|\mathbf{x}_t)$ in the denominator.
- Let's condition on $\mathbf{x}_0$ to make the reverse $q(\mathbf{x}_{t-1}|\mathbf{x}_t)$ tractable.

ELBO for Gaussian Diffusion Model

Bayes' Rule ($t \geq 2$)

$$q(\mathbf{x}_t|\mathbf{x}_{t-1}, \mathbf{x}_0) = \frac{q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)q(\mathbf{x}_t|\mathbf{x}_0)}{q(\mathbf{x}_{t-1}|\mathbf{x}_0)}$$

$$\mathcal{L}_{\phi,\theta}(\mathbf{x}) = \mathbb{E}_{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} \log \frac{p(\mathbf{x}_T)\prod_{t=1}^{T} p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t)}{\prod_{t=1}^{T} q(\mathbf{x}_t|\mathbf{x}_{t-1}, \mathbf{x}_0)}$$

$$= \mathbb{E}_{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} \log \frac{p(\mathbf{x}_T)p_{\theta}(\mathbf{x}_0|\mathbf{x}_1)\prod_{t=2}^{T} p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t)}{q(\mathbf{x}_1|\mathbf{x}_0)\prod_{t=2}^{T} q(\mathbf{x}_t|\mathbf{x}_{t-1}, \mathbf{x}_0)}$$

$$= \mathbb{E}_{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} \log \frac{p(\mathbf{x}_T)p_{\theta}(\mathbf{x}_0|\mathbf{x}_1)\prod_{t=2}^{T} p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t)}{q(\mathbf{x}_1|\mathbf{x}_0)\prod_{t=2}^{T} \frac{q(\mathbf{x}_{t-1}|\mathbf{x}_t,\mathbf{x}_0)q(\mathbf{x}_t|\mathbf{x}_0)}{q(\mathbf{x}_{t-1}|\mathbf{x}_0)}}$$

$$= \mathbb{E}_{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} \log \frac{p(\mathbf{x}_T)p_{\theta}(\mathbf{x}_0|\mathbf{x}_1)\prod_{t=2}^{T} p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t)}{q(\mathbf{x}_T|\mathbf{x}_0)\prod_{t=2}^{T} q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)}$$

Markov Property

$$q(\mathbf{x}_t|\mathbf{x}_{t-1}, \mathbf{x}_0) = q(\mathbf{x}_t|\mathbf{x}_{t-1})$$

Conditioning on $\mathbf{x}_0$ leaves the forward transitions unchanged.

Separate the decoder step $t = 1$.

Apply Bayes' rule for each $t \geq 2$.

Intermediate marginals telescope; the $q(\mathbf{x}_1|\mathbf{x}_0)$ factor cancels too.

ELBO for Gaussian Diffusion Model

Derivation (continued)

$$\mathcal{L}_{\phi,\theta}(\mathbf{x}) = \mathbb{E}_{q(\mathbf{x}_{1:T}|\mathbf{x}_0)}\Bigg[\log p_{\boldsymbol{\theta}}(\mathbf{x}_0|\mathbf{x}_1) + \log\frac{p(\mathbf{x}_T)}{q(\mathbf{x}_T|\mathbf{x}_0)} + \sum_{t=2}^{T}\log\left(\frac{p_{\boldsymbol{\theta}}(\mathbf{x}_{t-1}|\mathbf{x}_t)}{q(\mathbf{x}_{t-1}|\mathbf{x}_t,\mathbf{x}_0)}\right)\Bigg]$$

Expand the logarithm: products become sums.

$$= \mathbb{E}_{q(\mathbf{x}_1|\mathbf{x}_0)}\log p_{\boldsymbol{\theta}}(\mathbf{x}_0|\mathbf{x}_1) + \mathbb{E}_{q(\mathbf{x}_T|\mathbf{x}_0)}\log\frac{p(\mathbf{x}_T)}{q(\mathbf{x}_T|\mathbf{x}_0)} + \sum_{t=2}^{T}\mathbb{E}_{q(\mathbf{x}_{t-1},\mathbf{x}_t|\mathbf{x}_0)}\log\left(\frac{p_{\boldsymbol{\theta}}(\mathbf{x}_{t-1}|\mathbf{x}_t)}{q(\mathbf{x}_{t-1}|\mathbf{x}_t,\mathbf{x}_0)}\right)$$

Integrate out unused variables. Each term depends on one state or one adjacent pair.

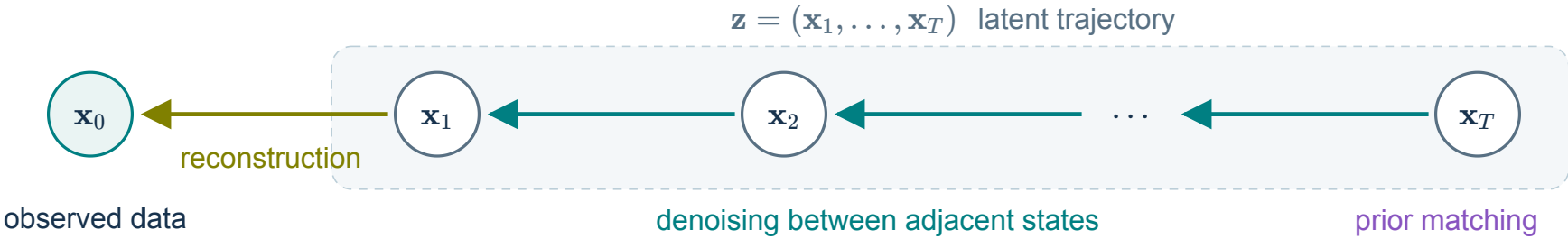
$$= \mathbb{E}_{q(\mathbf{x}_1|\mathbf{x}_0)}\log p_{\boldsymbol{\theta}}(\mathbf{x}_0|\mathbf{x}_1) - D_{\mathrm{KL}}\big(q(\mathbf{x}_T|\mathbf{x}_0)\|p(\mathbf{x}_T)\big) - \sum_{t=2}^{T}\underbrace{\mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)}D_{\mathrm{KL}}\big(q(\mathbf{x}_{t-1}|\mathbf{x}_t,\mathbf{x}_0)\|p_{\boldsymbol{\theta}}(\mathbf{x}_{t-1}|\mathbf{x}_t)\big)}_{\mathcal{L}_t}$$

Use the definition of KL. For each $\mathbf{x}_t$, average over $q(\mathbf{x}_{t-1}|\mathbf{x}_t,\mathbf{x}_0)$; then average over $q(\mathbf{x}_t|\mathbf{x}_0)$.

Ho J. Denoising Diffusion Probabilistic Models, 2020

ELBO for Gaussian Diffusion Model

$$\mathcal{L}_{\phi,\theta}(\mathbf{x}) = \mathbb{E}_{q(\mathbf{x}_1|\mathbf{x}_0)} \log p_{\boldsymbol{\theta}}(\mathbf{x}_0|\mathbf{x}_1) - D_{\mathrm{KL}}\big(q(\mathbf{x}_T|\mathbf{x}_0)\|p(\mathbf{x}_T)\big) - \sum_{t=2}^{T} \underbrace{\mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} D_{\mathrm{KL}}\big(q(\mathbf{x}_{t-1}|\mathbf{x}_t,\mathbf{x}_0)\|p_{\boldsymbol{\theta}}(\mathbf{x}_{t-1}|\mathbf{x}_t)\big)}_{\mathcal{L}_t}$$



First term — reconstruction: recover the clean image $\mathbf{x}_0$ from the slightly noisy $\mathbf{x}_1$.

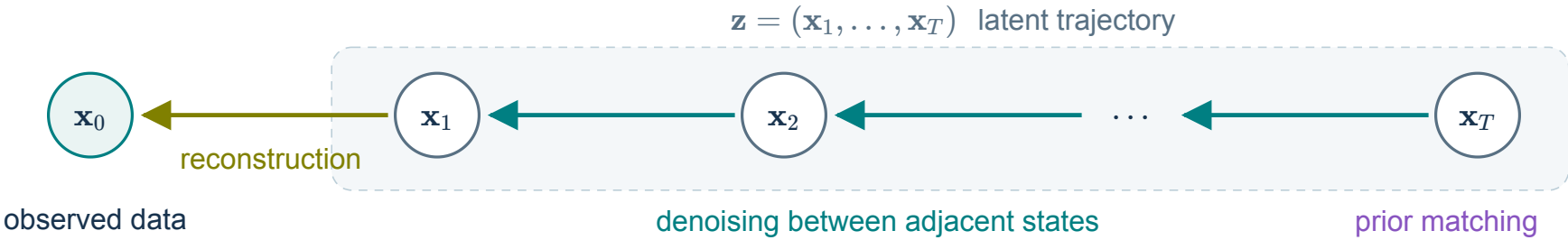
For a continuous decoder $p_{\boldsymbol{\theta}}(\mathbf{x}_0|\mathbf{x}_1) = \mathcal{N}\big(\boldsymbol{\mu}_{\boldsymbol{\theta},1}(\mathbf{x}_1), \sigma_1^2\mathbf{I}\big)$ with fixed $\sigma_1^2 > 0$:

$$-\mathbb{E}_{q(\mathbf{x}_1|\mathbf{x}_0)} \log p_{\boldsymbol{\theta}}(\mathbf{x}_0|\mathbf{x}_1) = \frac{1}{2\sigma_1^2} \mathbb{E}_{q(\mathbf{x}_1|\mathbf{x}_0)} \big\|\mathbf{x}_0 - \boldsymbol{\mu}_{\boldsymbol{\theta},1}(\mathbf{x}_1)\big\|^2 + \mathrm{const}.$$

Second term — prior matching: constant in $\boldsymbol{\theta}$, since q and $p(\mathbf{x}_T) = \mathcal{N}(0, \mathbf{I})$ are fixed.

ELBO for Gaussian Diffusion Model

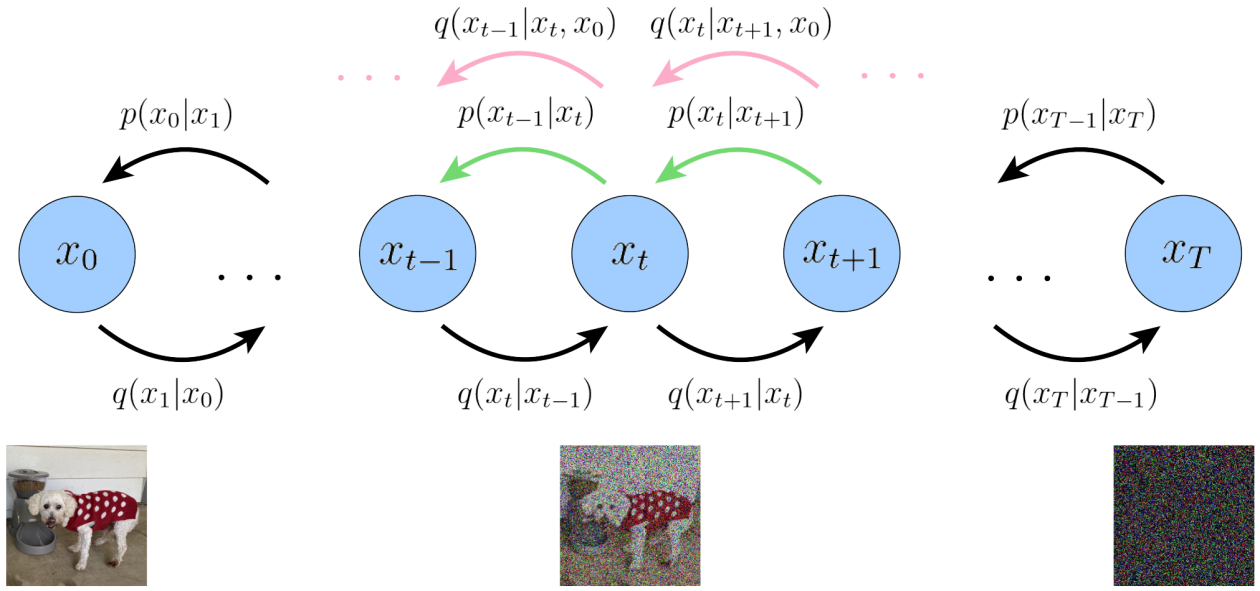
$$\mathcal{L}_{\phi,\theta}(\mathbf{x}) = \mathbb{E}_{q(\mathbf{x}_1|\mathbf{x}_0)} \log p_{\boldsymbol{\theta}}(\mathbf{x}_0|\mathbf{x}_1) - D_{\mathrm{KL}}\big(q(\mathbf{x}_T|\mathbf{x}_0)\|p(\mathbf{x}_T)\big) - \sum_{t=2}^{T} \underbrace{\mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} D_{\mathrm{KL}}\big(q(\mathbf{x}_{t-1}|\mathbf{x}_t,\mathbf{x}_0)\|p_{\boldsymbol{\theta}}(\mathbf{x}_{t-1}|\mathbf{x}_t)\big)}_{\mathcal{L}_t}$$



Third term — intermediate denoising: match each learned reverse step to its conditioned target. During training, knowing $\mathbf{x}_0$ makes this target distribution tractable.

$$\mathcal{L}_t = \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} D_{\mathrm{KL}}\Big(\underbrace{q(\mathbf{x}_{t-1}|\mathbf{x}_t,\mathbf{x}_0)}_{\text{known target}} \Big\| \underbrace{p_{\boldsymbol{\theta}}(\mathbf{x}_{t-1}|\mathbf{x}_t)}_{\text{learned reverse step}}\Big), \qquad t \geq 2.$$

Intermediate Denoising: KL Between Gaussians



Known Gaussian Target

$$q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \mathcal{N}\left(\tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t, \mathbf{x}_0), \tilde{\beta}_t \mathbf{I}\right)$$

Learned Gaussian Reverse Step

$$p_{\boldsymbol{\theta}}(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}\left(\boldsymbol{\mu}_{\boldsymbol{\theta},t}(\mathbf{x}_t), \boldsymbol{\sigma}^2_{\boldsymbol{\theta},t}(\mathbf{x}_t)\right)$$

$$\mathcal{L}_t = \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} D_{\mathrm{KL}}\left(\mathcal{N}\left(\tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t, \mathbf{x}_0), \tilde{\beta}_t \mathbf{I}\right) \middle\| \mathcal{N}\left(\boldsymbol{\mu}_{\boldsymbol{\theta},t}(\mathbf{x}_t), \boldsymbol{\sigma}^2_{\boldsymbol{\theta},t}(\mathbf{x}_t)\right)\right), \quad t \geq 2.$$

For each $\mathbf{x}_t$, the **KL between these Gaussians has a closed form**.

From Gaussian KL to Mean Regression

Known Gaussian Target

$$q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \mathcal{N}\left(\tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t, \mathbf{x}_0), \tilde{\beta}_t \mathbf{I}\right)$$

Learned Gaussian Reverse Step

$$p_{\boldsymbol{\theta}}(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}\left(\boldsymbol{\mu}_{\boldsymbol{\theta},t}(\mathbf{x}_t), \boldsymbol{\sigma}^2_{\boldsymbol{\theta},t}(\mathbf{x}_t)\right)$$

Fix the Covariance, Learn the Mean

Using the known target covariance simplifies learning: the network only needs to predict the mean.

$$\boldsymbol{\sigma}^2_{\boldsymbol{\theta},t}(\mathbf{x}_t) = \tilde{\beta}_t \mathbf{I}, \qquad t \geq 2.$$

The noise schedule fixes $\tilde{\beta}_t$; the network predicts the mean.

$$\begin{aligned}
\mathcal{L}_t &= \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} D_{\mathrm{KL}}\left(\mathcal{N}\left(\tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t, \mathbf{x}_0), \tilde{\beta}_t \mathbf{I}\right) \| \mathcal{N}\left(\boldsymbol{\mu}_{\boldsymbol{\theta},t}(\mathbf{x}_t), \tilde{\beta}_t \mathbf{I}\right)\right) \\
&= \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} \left[\frac{1}{2\tilde{\beta}_t} \left\| \underbrace{\tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t, \mathbf{x}_0)}_{\text{known target}} - \underbrace{\boldsymbol{\mu}_{\boldsymbol{\theta},t}(\mathbf{x}_t)}_{\text{network prediction}} \right\|^2 \right].
\end{aligned}$$

Same covariance → weighted regression onto the target mean.

Training and Sampling

$$-\mathcal{L}_{\boldsymbol{\phi},\boldsymbol{\theta}}(\mathbf{x}_0) = -\mathbb{E}_{q(\mathbf{x}_1|\mathbf{x}_0)} \log p_{\boldsymbol{\theta}}(\mathbf{x}_0|\mathbf{x}_1) + D_{\mathrm{KL}}\big(q(\mathbf{x}_T|\mathbf{x}_0)\|p(\mathbf{x}_T)\big) + \sum_{t=2}^{T} \mathcal{L}_t.$$

Training

1. Sample $\mathbf{x}_0 \sim p_{\text{data}}$, $t \sim \text{Uniform}\{2, \dots, T\}$.
2. Sample $\boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I})$; construct $\mathbf{x}_1$, $\mathbf{x}_t$.
3. Minimize $\hat{\mathcal{L}}$ by SGD: the omitted prior KL is independent of $\boldsymbol{\theta}$.

$$\mathbf{x}_s = \sqrt{\bar{\alpha}_s}\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_s}\boldsymbol{\epsilon}, \quad s \in \{1, t\}.$$

$$\hat{\mathcal{L}} = -\log p_{\boldsymbol{\theta}}(\mathbf{x}_0|\mathbf{x}_1) + \frac{T-1}{2\tilde{\beta}_t}\left\|\tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t, \mathbf{x}_0) - \boldsymbol{\mu}_{\boldsymbol{\theta},t}(\mathbf{x}_t)\right\|^2.$$

$T-1$: uniform timestep average → sum.

Sampling

1. Sample $\mathbf{x}_T \sim \mathcal{N}(0, \mathbf{I})$.
2. Denoise $\mathbf{x}_{t-1} = \boldsymbol{\mu}_{\boldsymbol{\theta},t}(\mathbf{x}_t) + \sqrt{\tilde{\beta}_t}\boldsymbol{\epsilon}_t$.
3. Return the mean $\hat{\mathbf{x}}_0 = \boldsymbol{\mu}_{\boldsymbol{\theta},1}(\mathbf{x}_1)$.

For $t = T, \dots, 2$, sample the **reverse kernel**:

$$p_{\boldsymbol{\theta}}(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}\big(\boldsymbol{\mu}_{\boldsymbol{\theta},t}(\mathbf{x}_t), \tilde{\beta}_t\mathbf{I}\big).$$

Use fresh $\boldsymbol{\epsilon}_t \sim \mathcal{N}(0, \mathbf{I})$ at each step.

Summary

- Forward diffusion has exact Gaussian conditionals; a suitable schedule removes the signal and controls SNR.
- For small steps, Gaussian reverse transitions approximate the path from noise to data.
- Conditioning on the clean image gives a tractable Gaussian reverse distribution.
- Gaussian diffusion is a VAE with a hierarchy of latent variables and a fixed encoder.
- Its ELBO decomposes into reconstruction, prior matching, and denoising terms; Gaussian denoising terms reduce to squared error.